

HDOC - Day 10 Activity

C Programming Assignment

Aim

To prepare flowcharts, algorithms, pseudocode, test cases, C programs, compilation commands, execution results, and testing for the problems given in HDOC.

Program 1:

Write a C program to input time in seconds and convert it into days, hours, minutes, and seconds format.

Algorithm

1. Start.
2. Input total seconds.
3. If seconds are negative, display an error and stop.
4. Calculate days = total / 86400.
5. Find remaining seconds using total % 86400.
6. Calculate hours = remaining / 3600 and update remainder.
7. Calculate minutes = remaining / 60.
8. Remaining value is seconds.
9. Display days, hours, minutes and seconds.
10. Stop.

Pseudocode

START

INPUT total_seconds

IF total_seconds < 0 THEN DISPLAY "Invalid input"; STOP

days $\leftarrow$ total_seconds / 86400

total_seconds $\leftarrow$ total_seconds MOD 86400

hours $\leftarrow$ total_seconds / 3600

total_seconds $\leftarrow$ total_seconds MOD 3600

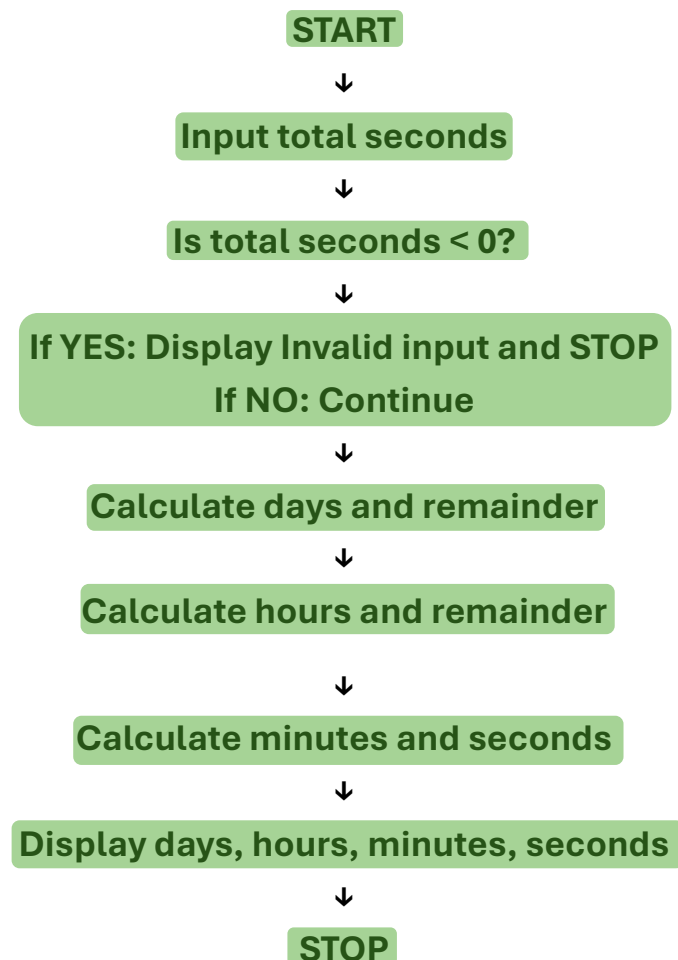
minutes $\leftarrow$ total_seconds / 60

seconds $\leftarrow$ total_seconds MOD 60

DISPLAY days, hours, minutes, seconds

STOP

Flowchart



Standard Test Case Table

Case	Input(s)	Expected output	Result
1	60	0 days, 0 hours, 1 minute, 0 second	Pass
2	3661	0 days, 1 hour, 1 minute, 1 second	Pass
3	90061	1 day, 1 hour, 1 minute, 1 second	Pass
4	-5	Invalid input!	Pass

C Program

```
#include <stdio.h>

int main() {
    long total, days, hours, minutes, seconds;
    printf("Enter time in seconds: ");
    scanf("%ld", &total);
    if (total < 0) {
        printf("Invalid input!\n");
        return 0;
    }
    days = total / 86400;
    total = total % 86400;
    hours = total / 3600;
    total = total % 3600;
    minutes = total / 60;
    seconds = total % 60;
    printf("%ld days, %ld hours, %ld minutes, %ld seconds\n",
        days, hours, minutes, seconds);
    return 0;
}
```

Compilation, Execution and Testing

```
Session Actions Edit View Help
(kali㉿kali)-[~]
$ vi time.c
(kali㉿kali)-[~]
$ cc time.c
(kali㉿kali)-[~]
$ ./a.out
Enter time in seconds: 60 time (years): 10000 5 2
0 days, 0 hours, 1 minutes, 0 seconds
(kali㉿kali)-[~]
$ ./a.out
Enter time in seconds: 3661
0 days, 1 hours, 1 minutes, 1 seconds
(kali㉿kali)-[~]
$ ./a.out
Enter time in seconds: 90061
1 days, 1 hours, 1 minutes, 1 seconds
(kali㉿kali)-[~]
$ ./a.out
Enter time in seconds: -5
Invalid input!
```

Result: The program successfully converts the entered time in seconds into days, hours, minutes and seconds.

Program 2:

Write a program to input a year and check whether it is a leap year or not.

Leap Year Rule

A year is a leap year if it is divisible by 400, or if it is divisible by 4 but not divisible by 100. For example, 2000 and 2024 are leap years, while 1900 is not.

Algorithm

1. Start.
2. Input year.
3. If year is less than or equal to 0, display invalid year and stop.
4. If year is divisible by 400, it is a leap year.
5. Otherwise, if year is divisible by 4 and not divisible by 100, it is a leap year.
6. Otherwise, it is not a leap year.

7. Display the result.

8. Stop.

Pseudocode

START

INPUT year

IF year ≤ 0 THEN DISPLAY "Invalid year"; STOP

IF (year MOD 400 = 0) OR ((year MOD 4 = 0) AND (year MOD 100 $\neq$ 0)) THEN
DISPLAY "Leap year"

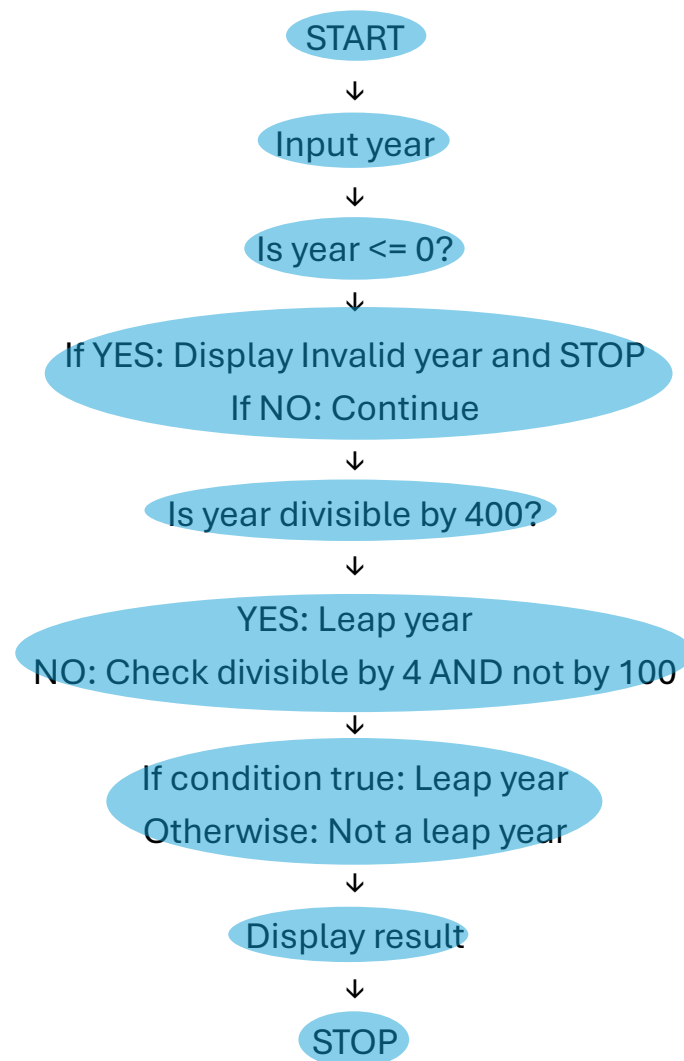
ELSE

DISPLAY "Not a leap year"

END IF

STOP

Flowchart



Standard Test Case Table

Case	Input(year)	Expected output	Result
1	2024	2024 is a leap year.	Pass
2	2023	2023 is not a leap year.	Pass
3	1900	1900 is not a leap year.	Pass
4	2000	2000 is a leap year.	Pass
5	-1	Invalid year!	Pass

C Program

```
#include <stdio.h>

int main() {
    int year;
    printf("Enter a year: ");
    scanf("%d", &year);
    if (year <= 0) {
        printf("Invalid year!\n");
        return 0;
    }
    if ((year % 400 == 0) ||
        (year % 4 == 0 && year % 100 != 0))
        printf("%d is a leap year.\n", year);
    else
        printf("%d is not a leap year.\n", year);
    return 0;
}
```

Compilation, Execution and Testing

```
(kali㉿kali) [~]
$ vi year.c

(kali㉿kali)-[~]
$ cc year.c

(kali㉿kali)-[~]
$ ./a.out
Enter a year: 2024nds: 60
2024 is a leap year.minutes: 0 seconds

(kali㉿kali)-[~]
$ ./a.out
Enter a year: 2023nds: 3661
2023 is not a leap year.es: 1 seconds

(kali㉿kali)-[~]
$ ./a.out
Enter a year: 1900nds: 90061
1900 is not a leap year.es: 1 seconds

(kali㉿kali)-[~]
$ ./a.out
Enter a year: 2000nds: -5
2000 is a leap year.

(kali㉿kali)-[~]
$ ./a.out
Enter a year: -1
Invalid year!
```

Result: The program successfully checks the entered year and correctly identifies whether it is a leap year or not.